Absorbance and Concentration · One complete topic

Short notes and keywords

Short answers, a keyword index, rapid recall and a last-10-minutes page.

Exactly the definitions, procedure, numerical or code example and limitations in the included lesson · compiled 17 October 2026

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1 Short notes

1 · Absorbance and Concentration

Absorbance measures attenuation on a logarithmic scale. In the linear Beer–Lambert range it is proportional to concentration and path length. This lesson keeps fractions, percentages and physical units distinct so that the algebra matches the measurement.

S1

2 Keyword index

2 terms; the number in brackets is the topic.

Absorbance (1). Negative base-ten logarithm of fractional transmittance. S1

Molar absorption coefficient (1). Wavelength- and species-dependent proportionality coefficient. S1

3 Rapid recall

1	Why must fractional transmittance be used in the logarithm?	The definition uses I divided by I0. A percentage is one hundred times that ratio and would shift the logarithm by two. S1

4 Last 10 minutes before the exam

Top facts

1. Use the fractional transmittance in the logarithm. The illustrative coefficient is not a universal value for copper(II). S1

Top formulas

E1.1 · Beer–Lambert relation

$$A = \varepsilon\, l\, c = -\log_{10} T$$

gives	Absorbance of a dilute absorbing species, such as illustrative $\mathrm{Cu^{2+}}$ in solution.
applies	Monochromatic measurement in the linear range with a suitable blank and consistent units.
units	A: dimensionless; ε: $\mathrm{L\,mol^{-1}\,cm^{-1}}$; l: cm; c: $\mathrm{mol\,L^{-1}}$.

S1